Binya Benson Abe Boro
MA Development Economics and International Studies, FAU Erlangen – Nürnberg, Germany

1. Introduction

Is public expenditure on education a topic of economic interest? Globally, improving the quality of education is a top priority for the 2030 agenda for sustainable development (United Nations 2015). According to the 2024 Education Finance Watch report, governments and households spent about US$55 per a school child in 2022. In addition, donor countries donated a record US$16.6 billion in 2022 for education in developing countries (World Bank 2024). This analysis pursued the question of whether or not spending on education improves the academic performance of pupils.

Whether public expenditure on education produces the desired effects for the pupil and the nation is of fundamental interest in economics. Already, economists have conducted both theoretical and empirical analysis of this relationship, giving birth to what is now known as the economics of education. One core theory is the human capital theory, which takes education as an investment in human capital (Becker (1962). The theory stipulates that investing in education generates the skills, knowledge, and competencies needed to increase individual and national productivity and earnings. Empirical analysis of this theory proved it right. Krueger and Lindahl (2001) resolved the micro and macro inconsistency of the effect of investing in education. Previous studies had shown positive effects on individual wages, but weak or no effects on economic growth. They resolved this by using IV to address a measurement error, finding strong positive and statistically significant effects on both wages and national growth. Using a Regression Discontinuity Design, Baron (2022) found that increasing annual school spending by US$1000 per pupil increased student test scores by about 8 percent and a reduction of dropout rate by 9 percent in poor districts of Wisconsin State. A similar study by Hyman (2017) examined the long-run effects of school spending in Michigan using 2SLS, with the State's school finance reform as a source of exogenous variation. He found that students exposed to a 10 percent increase in spending were 7 percent more likely to enrol in college and 11 percent more likely to earn a postsecondary degree.

Thus, the topic of school spending and student's performance is of a fundamental economic interest. This is because both economic theory and empirical evidence show that spending positively affects human capital formation, which in turn leads to growth in individual and national income. Studying this relationship ensures efficient allocation of education finance in the face of financial scarcity.

2. Descriptive Statistics

Table I below shows that the average test score of a pupil is 654 points, with the best pupil scoring 707 points and worst 606 points. The average expenditure per pupil in each district is US$5,312. The maximum expenditure per pupil is US$7,711 and the minimum is US$3,926. The average district has an income of US$15,317, with the poorest district having an income of US$5,335 while the richest US$55,328. Some districts have 100

percent of their pupils receiving food vouchers while others have none. On average, each pupil has at least one computer. However, some pupils have no access to computers even though some schools have at least four computers per pupil. The average class has about 16 percent share of non-native English speakers but some schools zero number of non-native English speakers.

Table 1: Summary Statistics

	Mean	SD	Minimum	Maximum	Observations
Average Test Scores	654.1565	19.05335	605.55	706.75	420
Expenditures Per Pupil	5312.408	633.9371	3926.07	7711.507	420
Share of Pupils with Food Vouchers (in %)	44.70524	27.12338	0	100	420
Number of Computers Per Pupil	.1359266	.0649558	0	.4208333	420
Average Income (in 1000 US$)	15.31659	7.22589	5.335	55.328	420
Share of Non-native English Speakers (in %)	15.76816	18.28593	0	85.53972	420

Figure I shows a histogram of the distribution of the average test scores. The shape is approximately symmetric, indicating that pupil's test scores are evenly distributed around the mean.

Figure I: Distribution of Pupil's Average Test Scores

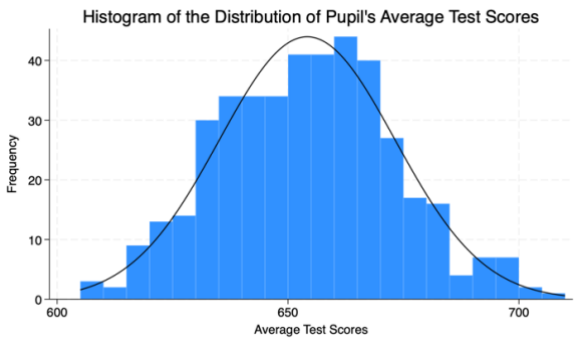


Figure II below shows a right-skewed distribution of expenditure per pupil. Thus, expenditure per pupil is evenly distributed around the mean.

Figure II: The Distribution of Public School Expenditure Per Pupil

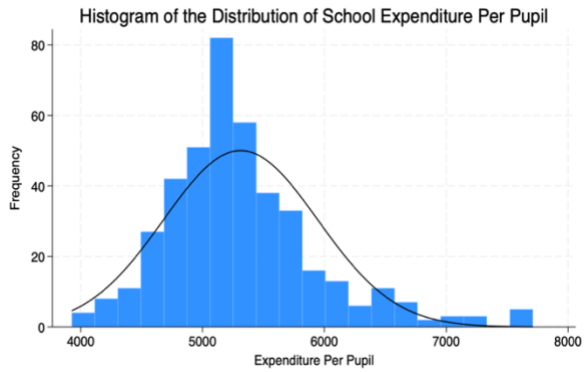
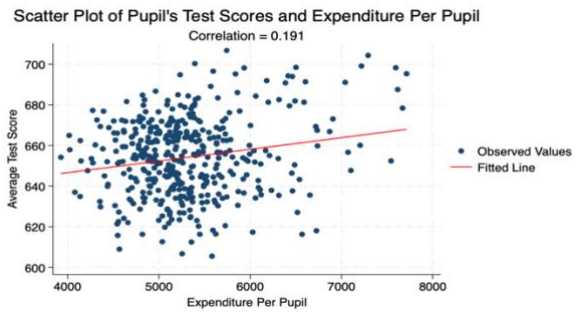


Figure III below shows a scatter plot of the relationship between public expenditure and pupil's test scores. The correlation coefficient of 0.191 indicates that their relationship is positive.

Figure III: The Relationship Between Public School Expenditure and Pupil's Test Scores



3. Data Analysis and Interpretation

3.1. Baseline Specification

This analysis examined whether public school expenditure improves pupil's performance. Thus, the key question is: Does public-school spending improve the performance of pupils?

3.2. Economic Model

Pupil's performance, as measured by average test scores, is a function of public-school expenditure, measured by expenditure per pupil, and covariates such as the share of pupils with food vouchers, number of computers per pupil, average income of the district, and share of non-native English speakers.

Average Test Scores = f (Expenditure Per Pupil, Covariates)

3.3. Econometric Model

$$y_i = \beta_0 + \beta_1 x + \gamma \mathbf{X} + \varepsilon_i$$

Where: $\mathbf{y_i}$ is the dependent variable, average test scores; x is the main independent variable, expenditure per pupil; $\mathbf{X}$ is a vector of covariates; and $\boldsymbol{\varepsilon_i}$ is the disturbance term capturing unobserved factors. The coefficient of interest, $\boldsymbol{\beta_1}$, captures the effect of public-school expenditure on pupil's average test scores.

3.4. OLS Estimation of the Baseline Model

Table II shows the OLS regression results. The first column is a bivariate regression of pupil's average test scores on expenditure per pupil. The coefficient is large, positive, and statistically significant. However, the analysis focuses on the second column, which shows estimates of the multivariate baseline model. The coefficient of interest (β_1 = 8.906) implies that a one percent increase in expenditure per pupil is associated with 0.089 points (8.906/100) increase in pupil's average test scores, holding other factors constant. This is statistically significant at the 5% level (*p-value* <0.05). The model explains about 81 percent of the variation in pupil's average test scores (R_Squared = 0.808) and is jointly significant (*F-statistic p-value < 0.001)*. On average, one percentage point increase in the share of non-native English speakers is associated with 0.187 points decrease in pupil's average test scores, ceteris paribus. This is significant at the 1% level. Increasing the number of computers per pupil by one computer on average increases pupil's average test scores by 13.9 points, ceteris paribus. Increasing the income of the district by one dollar on average increases pupil's average test scores by 0.624 points, holding other factors constant. This is also significant at the 1% level. Finally, one percentage point increase in the share of pupils with food vouchers is associated with a decrease of 0.406 points in pupil's average test scores, other factors constant. This is significant at the 1% level. This result implies that public school expenditure does improve pupil's performance.

Table II: OLS Regression of Average Test Scores on Expenditure Per Pupil

	(1) Bivariate Regression	(2) Multivariate Regression
log of expenditure per pupil	29.465***	8.906**
	(8.008)	(4.016)
Share of non-native English speakers (in %)		-0.187***
		(0.031)
Number of computers per pupil		13.887**
		(6.779)
Average income (in 1000 US$)		0.624***
		(0.088)
Share of pupils with food vouchers (in %)		-0.406***
		(0.028)
Constant	401.607***	587.459***
	(68.642)	(33.492)
F_Statistic	13.539	348.486
Prop > F	0.000	0.000
R_Squared	0.031	0.808
Adjusted R_squared	0.029	0.806
Observations	420	420

Standard errors in parentheses
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

4. Functional Form and Model Testing

Table III column one below shows that the baseline model is better than the second model in column two. This is because the Adjusted R-squared of the baseline model (0.806) is larger than that of the second model (0.798). Also, the baseline model has

a lower AIC (2985) and BIC (3009) as compared to the second model's AIC (3001) and BIC (3025). Therefore, the baseline specification can better explain the relationship between public school expenditure and pupil's performance.

Table III: Testing and Comparing Different Model Specifications

	(1) Baseline Model	(2) Log-linear Model
log (expenditure per pupil)	8.906**	11.735***
	(4.016)	(4.030)
Share of non-native English speakers (in %)	-0.187***	-0.168***
	(0.031)	(0.032)
Number of computers per pupil	13.887**	16.932**
	(6.779)	(6.913)
Average income (in 1000 US$)	0.624***	
	(0.088)	
Share of pupils with food vouchers (in %)	-0.406***	-0.410***
	(0.028)	(0.031)
log (average income)		10.492***
		(1.806)
Constant	587.459***	544.488***
	(33.492)	(32.818)
Adjusted R_squared	0.806	0.798
AIC	2985.447	3001.159
BIC	3009.688	3025.400
Observations	420	420

Standard errors in parentheses
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

5. Analysis of Heteroskedasticity in the baseline model.

5.1. White Test

Table IV shows OLS estimation of the baseline model with and without robust standard errors. It also includes the *p-value* of the white test for heteroskedasticity. Since the white test *p-value (0.734) is greater than* 0.050, homoskedasticity is not rejected at the 5% level, implying no evidence of heteroskedasticity. This implies that the explanatory variables are uncorrelated with the disturbance term. In column 2, I report OLE estimates with robust standard errors. Unlike the standard errors, the coefficients remain similar.

Table IV: OLS Regression with Robust Standard Errors

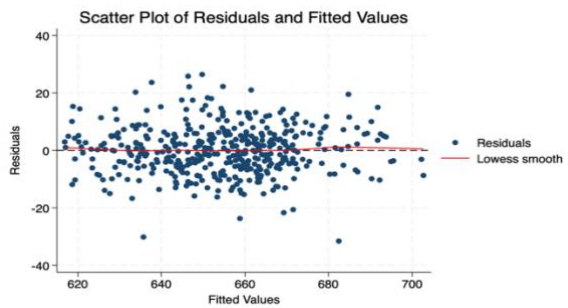
	(1) OLS	(2) OLS with Robust SE
log (expenditure per pupil)	8.906**	8.906**
	(4.016)	(4.257)
Share of non-native English speakers (in %)	-0.187***	-0.187***
	(0.0313)	(0.0321)
Number of computers per pupil	13.89**	13.89*
	(6.779)	(7.558)
Average income (in 1000 US$)	0.624***	0.624***
	(0.0875)	(0.0841)
Share of pupils with food vouchers (in %)	-0.406***	-0.406***
	(0.0280)	(0.0299)
Constant	587.5***	587.5***
	(33.49)	(35.53)
Observations	420	420
R_Squared	0.808	0.808
White Test p_value	0.734	0.734

Standard errors in parentheses
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

5.2. Graphical Analysis of heteroskedasticity

Figure IV below shows a scatter plot of residuals against fitted values. The plot shows no distinct patterns, implying no evidence of heteroskedasticity. In addition, the lowess smooth line is approximately flat and centered around zero, which further suggests no evidence of heteroskedasticity.

Figure IV: Scatter Plot of Residuals and Fitted Values



6. Endogeneity

So far, the tests above reveal that the baseline model can better explain the effect of public expenditure on pupil's performance. The model is also homoscedastic, meaning that the OLS estimates are the best linear unbiased estimates. Furthermore, a Ramsey RESET test, analysis only shown in the dofile, fail to reject the null hypothesis that the baseline model has no omitted variables. However, this does not imply that the effect of

public expenditure on pupil's performance estimated here is causal. This is because several unobserved factors might still influence either educational expenditure or pupil's performance or both.

One unobserved factor is parental level of education. Highly educated parents are more likely to demand excellent academic performance from their children than those who are less educated. Another factor is parental level of income, which might influence both educational outcomes of children and public expenditure per pupil. Rich parents are more likely to not need external funding for their children and send their children to high performing schools than poor parents. Likewise, governments are more likely to spend more on poor schools and children than on rich schools. However, how pupils perform might also determine continuity of funding. These factors rise a possibility of omitted variables and endogeneity problems in the baseline model. Thus, the OLS estimates cannot be interpreted as causal.

7. Conclusion

This analysis pursued the question of whether public-school expenditure does improve pupil's performance or not. Analysis of a school spending data using OLS suggests that a one percent increase in expenditure per pupil on average increases pupils' performance by 0.089 points. Though this estimate is valid and consistent, there is still a possibility of endogeneity due to unobserved factors such as parental education, income, and policy information on how public expenditure is allocated per school and pupil. Thus, the OLE estimate cannot be interpreted as a causal effect of public expenditure on pupil's performance.

References

Baron, E Jason. 2022. "School Spending and Student Outcomes: Evidence from Revenue Limit Elections in Wisconsin." *American Economic Journal: Economic Policy* 14 (1): 1-39

Becker, Gary S. "Investment in Human Capital: A Theoretical Analysis." Journal of Political Economy, no. 5, Part 2(1962): 9-49

Hyman, Joshua. 2017. "Does Money Matter in the Long Run? Effects of School Spending on Educational Attainment." *American Economic Journal: Economic Policy* 9(4): 256-80

Krueger, Alan B., and Mikael Lindahl. "Education for Growth: Why and For Whom?" *Journal of Economic Literature* 39, no. 4 (2001): 1101–36. http://www.jstor.org/stable/2698521.

United Nations, "Transforming Our World: The 2030 Agenda for Sustainable Development," Resolution adopted by the General Assembly, 2015

World Bank, "Education Finance," Education Finance Watch Report, 2024.